Chapter 2 Exploring Data with Tables and Graphs
Histograms (2.2)

Objectives

In this lesson, we will learn how to

- describe what a histogram is and how it represents quantitative data;
- distinguish between a frequency histogram and a relative frequency histogram;
- use histograms to study center, variation, distribution shape, and possible outliers;
- recognize common distribution shapes, including bell-shaped, uniform, skewed right, and skewed left;
- determine the direction of skewness from the long tail;
- use normal quantile plots to help assess whether data appear to come from a normal distribution.

1. Why Histograms Are Useful

A frequency distribution is useful for organizing and summarizing quantitative data.

However, a table does not always make the overall pattern easy to see.

A histogram turns the frequency distribution into a picture.

Key Idea

A histogram is often easier to interpret than a table of numbers because it visually displays the overall distribution of the data.

Raw Data → Frequency Distribution → Histogram

2. Histogram

Histogram

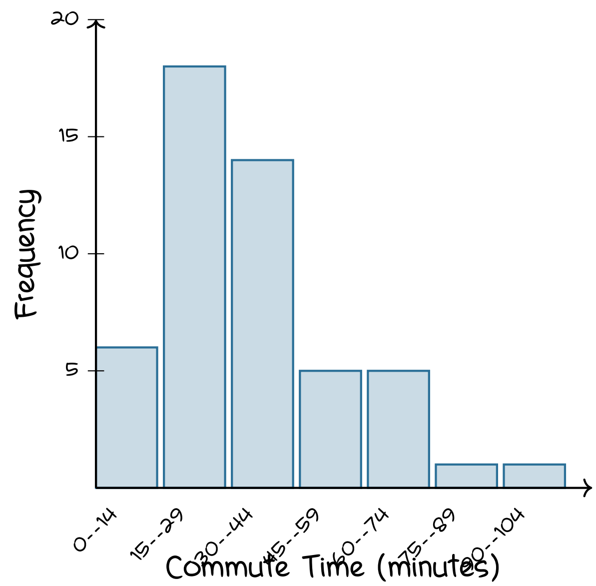
A histogram is a graph consisting of bars of equal width drawn adjacent to each other, unless there are gaps in the data.

- The horizontal scale represents classes of quantitative data values.
- The vertical scale represents frequencies.
- The height of each bar corresponds to the frequency of that class.
- Adjacent bars normally touch because neighboring classes represent consecutive intervals of quantitative values.

Consider the Los Angeles commute-time frequency distribution from Section 2.1:

Commute Time (minutes)	Frequency
$0 - 14$	6
$15 - 29$	18
$30 - 44$	14
$45 - 59$	5
$60 - 74$	5
$75 - 89$	1
$90 - 104$	1

The corresponding histogram is:



Important Uses of a Histogram

A histogram can help us identify:

- the shape of the distribution;
- the approximate center of the data;
- the spread or variation of the data;
- possible outliers.

Read the Histogram

Without doing any calculations, which commute-time class appears most common?

3. Relative Frequency Histogram

A histogram does not have to use raw frequencies on the vertical axis.
It can instead use proportions or percentages.

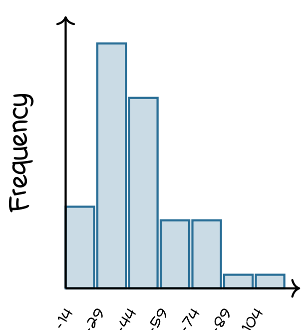
Relative Frequency Histogram

A relative frequency histogram has:

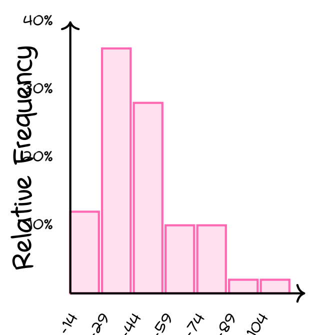
- the same classes;
- the same horizontal scale;
- the same overall shape as the corresponding frequency histogram;
- relative frequencies or percentages on the vertical scale instead of actual frequencies.

For example, if the class $15 - 29$ has frequency 18 out of 50 commute times,

$$\frac{18}{50} = 0.36 = 36\%.$$



Frequency Histogram



Relative Frequency Histogram

Very Important Observation

$\boxed{\text{Changing frequency to relative frequency does not change the shape.}}$

Only the vertical scale changes.

Example 1:
Suppose a data set contains 50 observations.
One class has frequency 14.
Find the relative frequency and percentage frequency for that class.

4. Critical Thinking: Interpreting Histograms

When examining a histogram, we want to do more than identify the tallest bar.

We want to ask what the graph tells us about the data.

A useful guide is CVDOT.

CVDOT

C Center: Where is the approximate middle of the data?

V Variation: How much are the values spread out?

D Distribution: What is the overall shape?

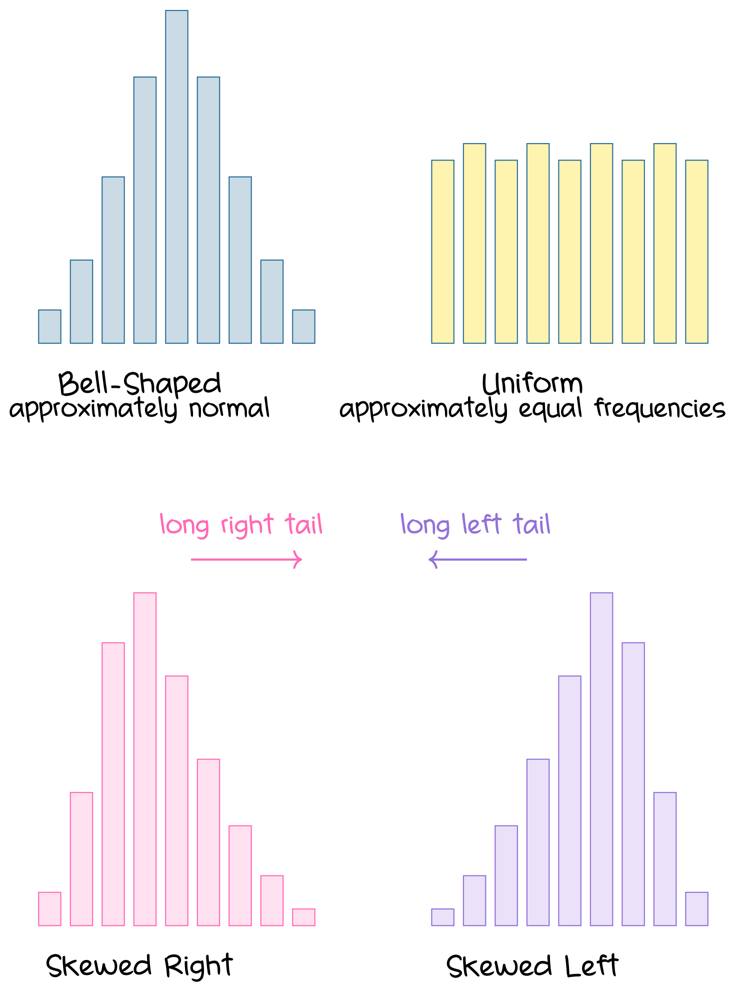
O Outliers: Are there unusual values far away from the rest?

T Time: If the data involve time, is there a changing pattern?

Example 2:
A histogram has one high region near the middle.
The bar heights decrease on both sides in a roughly symmetric pattern.
What shape does this suggest?

5. Common Distribution Shapes

Histograms can have several common shapes.



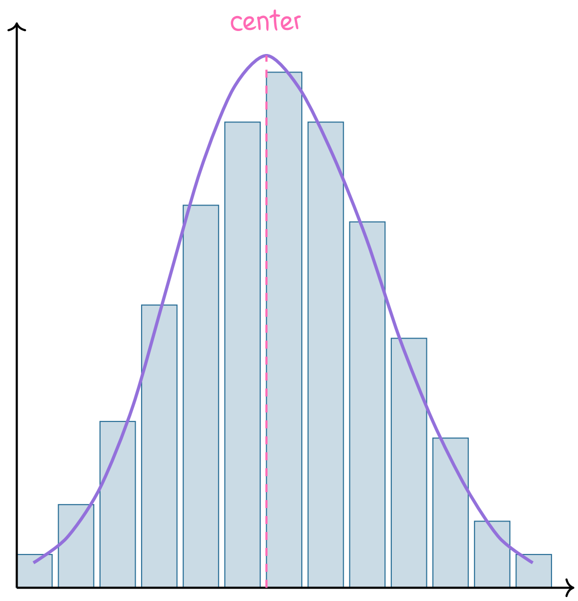
Bell-Shaped or Normal Distribution

Normal Distribution

A distribution may be approximately normal when its histogram is

- roughly bell-shaped;
- approximately symmetric;
- relatively high near the center;

- lower toward both ends.



roughly bell-shaped and symmetric

Visual Clue

For a roughly normal distribution,

$$\boxed{\text{low} \longrightarrow \text{high near center} \longrightarrow \text{low}}$$

and the two sides are approximately mirror images.

Uniform Distribution

Uniform Distribution

A uniform distribution has frequencies that are approximately the same throughout the distribution.
Its histogram therefore appears relatively flat.

A perfectly uniform distribution would have bars of exactly the same height, but real sample data usually show some small variation.

Skewness

Skewness

A distribution is skewed if it is not symmetric and extends farther on one side than on the other.

- Data skewed to the right, also called positively skewed, have a longer right tail.
- Data skewed to the left, also called negatively skewed, have a longer left tail.

Memory Tip

The direction of skewness is the direction of the long tail.

Do Not Follow the Peak

The direction of skewness is not determined by where most of the bars are located.
Always look at the direction in which the long, thin tail extends.

Example 3:
Most observations are concentrated on the left side of a histogram, but a small number of large observations stretch far to the right.
Is the distribution skewed left or skewed right?

6. Assessing Normality with Normal Quantile Plots

A histogram gives a useful visual impression of the shape of a distribution.
Another tool for assessing normality is the normal quantile plot.

Normal Quantile Plot

A normal quantile plot compares the observed sample data with the pattern we would expect if the population had a normal distribution.

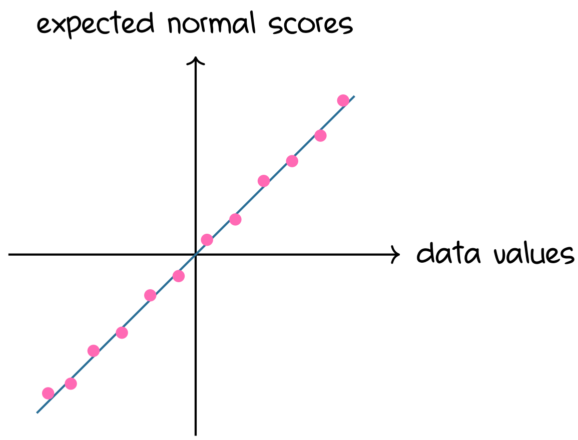
Criteria for Assessing Normality

Approximately Normal
The points are reasonably close to a straight-line pattern, and there is no other obvious systematic pattern.

Probably Not Normal
The population distribution is likely not normal if either of the following occurs:

- the points do not lie reasonably close to a straight line;
- the points show a systematic pattern that is not a straight-line pattern.

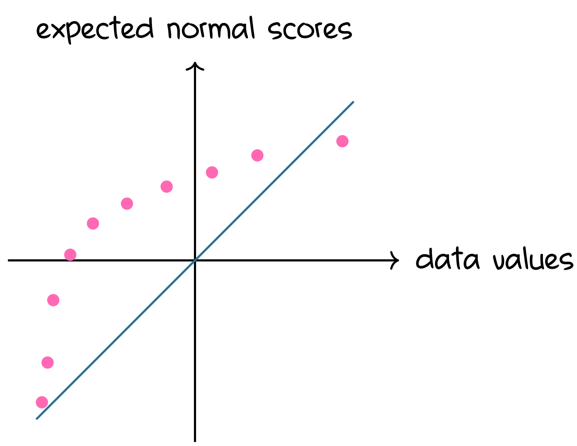
Case 1: Approximately Normal



Approximately Normal
points are reasonably close to a straight line

The points do not need to lie perfectly on the line.
The important question is whether the overall pattern is reasonably straight.

Case 2: Points Far from a Straight Line

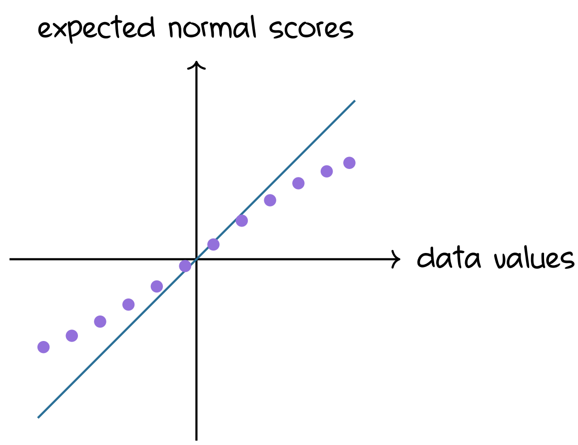


Not Normal
points are not reasonably close to a straight line

Here the points clearly fail to follow the straight-line pattern.

This suggests that the data do not come from a normal distribution.

Case 3: Systematic Curved Pattern



Not Normal
points show a systematic curved pattern

Even though some points may be close to the reference line, the overall curved pattern is systematic.

That also suggests a nonnormal distribution.

Important

A few small deviations from the reference line are not enough to conclude that a distribution is nonnormal.
We are looking for the

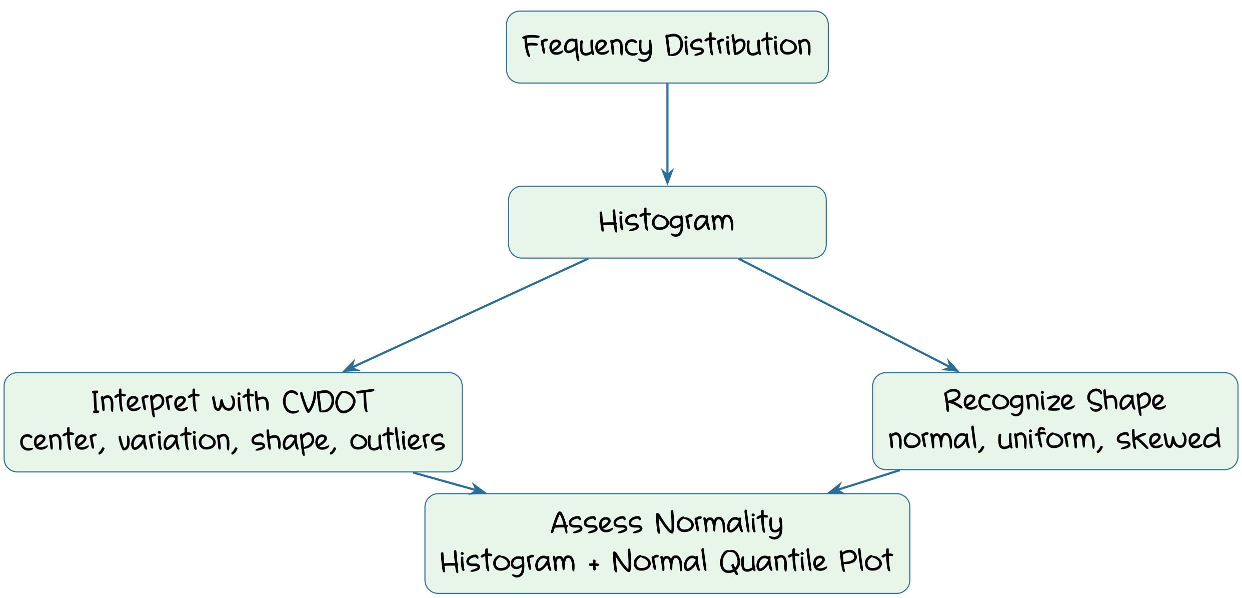
$\boxed{\text{overall pattern}}$

of the points.

Example 4:
A normal quantile plot has points that are fairly close to the line near the center, but the points systematically curve away from the line near the ends.

What should we conclude?

7. The Big Picture



Summary

Key Ideas

- A histogram displays quantitative data using adjacent bars.
- The horizontal scale represents classes of quantitative data values.
- The vertical scale represents frequency or relative frequency.
- A relative frequency histogram has the same shape and horizontal scale as the corresponding frequency histogram.
- Histograms help us study center, variation, distribution shape, outliers, and time.
- Common distribution shapes include:
  - bell-shaped or approximately normal;
  - uniform;
  - skewed right;
  - skewed left.
- A uniform distribution has approximately equal frequencies throughout the range.
- A right-skewed distribution has a long right tail.
- A left-skewed distribution has a long left tail.
- The direction of skewness is determined by the direction of the long tail.
- A normal quantile plot supports normality when the points are reasonably close to a straight line and show no other systematic pattern.
- A normal quantile plot suggests nonnormality when the points are far from a straight line or show a systematic nonlinear pattern.